Hey guys, welcome to my second tutorial, this one on Player Movement!
When we ended the last one, we had just created a square that would be our player in the game. Let's continue from there.

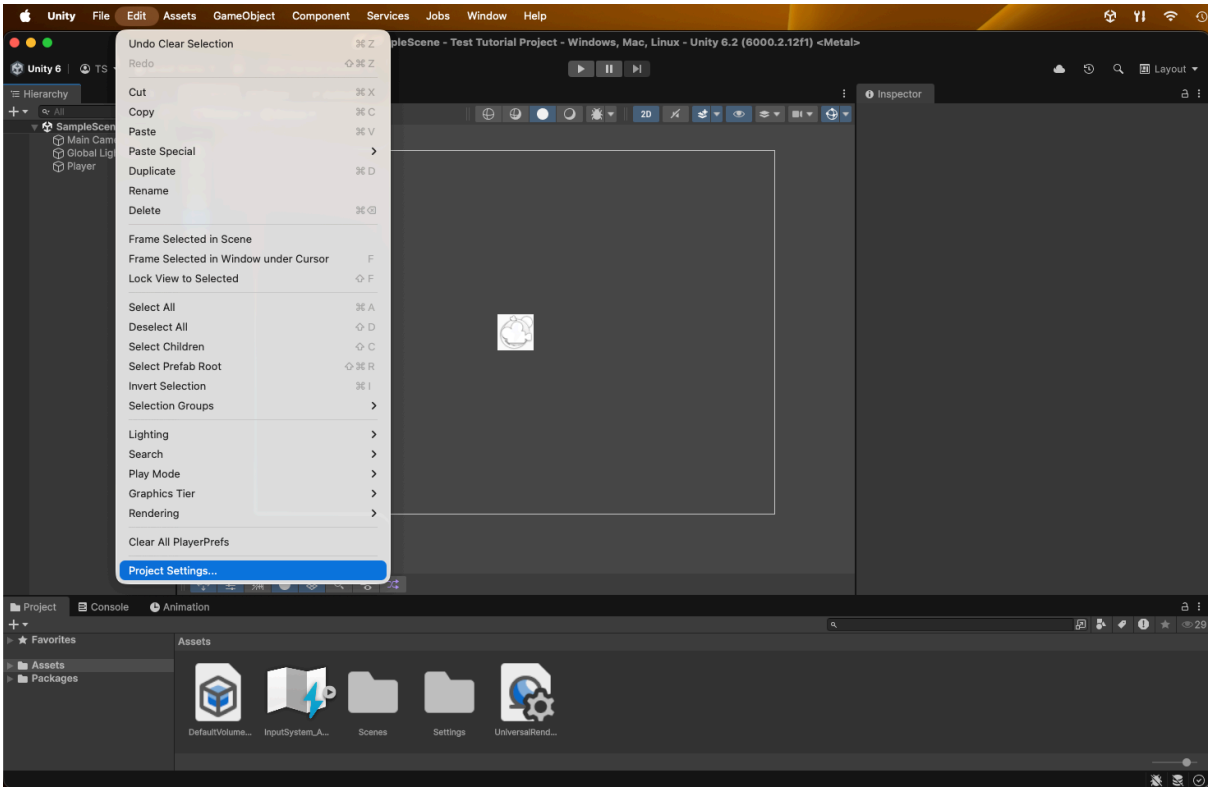
Now, before we start, no matter what you do in Unity, you will need to eventually look at scripts, if not write them. To do this, you need a type of app known as an IDE (Integrated Development Environment); basically an app that you code in. I would recommend installing Visual Studio Code here; it's free and newbie-friendly. Then, when that has been installed, open up Visual Studio Code, select the "Extensions" tab on the left (the one with the four squares as the icon), and search up and install the "Unity" extension. This will install three extensions that will allow you to find errors in your code whilst writing, and it will suggest possible keywords and functions as you are typing them to eliminate typos and speed up programming.

Now, I'm gonna teach you guys how to make your own player movement, but there may be some of you who don't want to deal with C# or programming in general. Understandable! You can definitely make a small game without knowing how to program, with the magic of packages. Packages are basically little bits of code, models, animations, or systems that Unity or other programmers have made and have put on the asset store for you to use. Some are free; some are not.

Of course, be careful downloading files off the internet. Asset Store is decently safe, but look for assets with good reviews. Packages are available on the Unity Asset Store, and some of the best ones are free! If you don't want to program (for some odd reason) go to this link: Asset Store, and follow the download instructions to get what is (probably) a better and more advanced system (I haven't used this package myself, so I'm not sure). However, the way I'm about to teach you is easier to understand.

There are currently two supported ways to collect input from the player's keyboard/joystick/mouse/whatever. The new one, which comes as the standard in versions past Unity 6, is a lot more complicated to set up and code with, but is more versatile in the long run. However, in the interest of simplicity, we will use the older, simpler input system. For information about using the new input system, go here: Unity Input Tutorial.

Ok; let's get started with the actual tutorial. When you have your project open, navigate to Edit > Project Settings in the top bar:

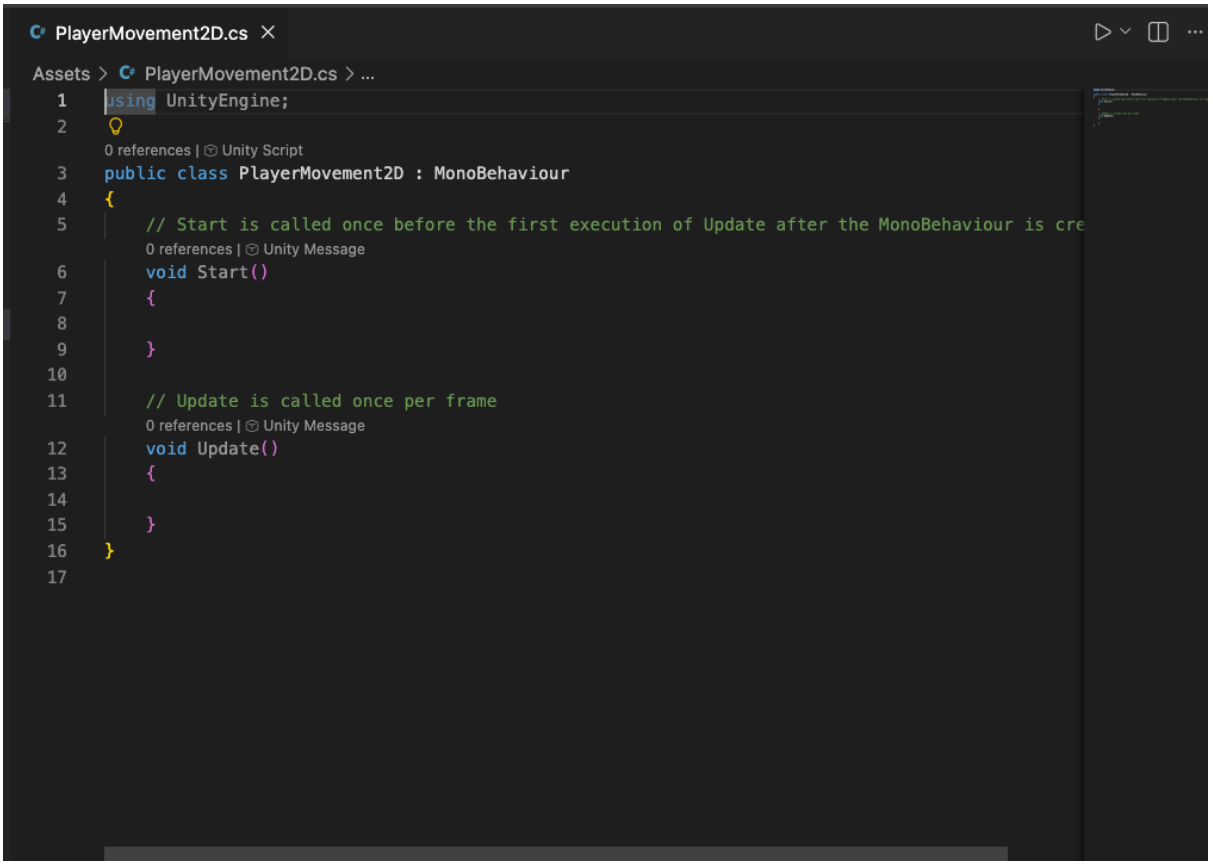


Now open the project settings, and navigate to “Player” in the left hand menu.
Type “Input” in the search bar, and scroll down in the “Player” menu to the highlighted word. It should read “Active Input Handling” and have a selection box next to it. Select the “Both” option, and then select Apply in the pop-up. Your editor will briefly restart, and you can now use the older input system in this project!

Go back to your Unity file, and click on the “Player”, or the square we created last tutorial in the Hierarchy. A bunch of BS should come up in the Inspector tab, showing the settings that we have on this object. We want to add a new component, so click on the “Add Component” button in the bottom of the Inspector tab. You may need to scroll down to find it. Then, search “Rigidbody 2D” and click on the component shown. A rigidbody is basically added to objects that you want to experience physics. If you now click Play (middle top of the screen), your square will now drop off the screen. This is because there’s no floor for it to contact. To stop them falling for now, stop the game, find the “Gravity Scale” setting in the Rigidbody component in the player’s Inspector, and set it to zero.

Now, we’re gonna make the actual script that will respond to input and make the player move. Go to the “Add Component” button in your inspector tab again, and this time, type “PlayerMovement2D”, and click on the “New Script” button, then click on “Create And Add”, and it should generate a popup confirming that you want to save the script to the assets folder. You’ve created your first script!

In your “Assets” section (bottom of the screen), you should see the “PlayerMovement2D” file. This file will have a white file with a big green hashtag on it as an icon. Select that file, and it should open up the file in Visual Studio Code (or whatever IDE you have used). Now, your script is gonna look something like this:



Now, we're going to enter in our code. Feel free to copy-paste what I've posted below, but please take the time to read the comments (in green) that I've written to show you how this script works (and C# in general), as it would really help you debug your scripts (and indeed those of the tutorials you will doubtless find online) in future.

```
// NOTE: This script uses the 'old' input system, which has not been standard since
Unity 6.
// Any tutorials from after 2025 or so will likely not use this input system.

// this is where we import packages. Packages are basically bits of code written by
other programmers (in this case Unity) that we use to define and use more advanced
bits of code.
using System.Collections;
using System.Collections.Generic;
using UnityEngine;
public class PlayerMovement2D : MonoBehaviour
{
   // this section of the script is where the different variables are initialised.
   // To initialise variables in C#, you write <access level> <variable type>
<variable name> = <variable value>, as shown below:
   private Rigidbody2D player = null; // The rigidbody is basically the component
that lets the game object it is attached to experience physics, e.g gravity,
bouncing off other objects, etc.
```

```
    private float moveSpeed = 10f;
    // The Start function is called when this script is activated for the first
time, e.g upon the start of the game.
    void Start()
    {
        // this code is basically just assigning values to the variable that we
previously left as null.
        player = GetComponent<Rigidbody2D>();
    }
    // this Update function is called every frame, making it the main function you
will code in.
    void Update()
    { // this line of code takes input from the horizontal axis (A and D keys) and
uses it to set the player's velocity via the rigidbody component.
        player.linearVelocity = new Vector2(Input.GetAxis("Horizontal") * moveSpeed,
player.linearVelocity.y);
    }
}
```

Once you've pasted that in, go back to your Unity project. It will take a moment for the changes in the script to be registered, as signalled by the loading bar that will come up. Once that's done, click play. Your square should move from side to side when you press the "A" and "D" keys.

That's it for this basic player movement tutorial. If you have any questions, please contact me via the discord server. Good luck!